

Common-Mode Voltage Reduction in Two-Level Inverters by Introducing Adaptive Carrier-Phase-Shift Method

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Abstract—Two-level voltage source inverters (2L-VSIs) are widely used in industrial motor drives due to their simplicity and cost-effectiveness. However, their pulsating output voltages introduce common-mode voltage (CMV) issues, leading to challenges such as electromagnetic interference (EMI), winding insulation stress, and bearing currents. While adopting an advanced topology or employing common-mode filters can mitigate CMV-based problems, these solutions often increase system complexity and cost. Alternatively, modulation-based techniques offer a simpler, hardware-free approach to reduce CMV. This paper proposes a modulation-based technique that introduces an adaptive carrier phase shift for sinusoidal pulse width modulation (SPWM) to address CMV issues in 2L-VSIs. Unlike conventional methods that rely on fixed carrier phase shifts (e.g., interleaved 0°, 120°, and 240°), the proposed approach dynamically adjusts the phase shift within each switching interval to effectively suppress CMV. The proposed technique was validated through both experimental testing and analytical modeling to demonstrate CMV reduction. This approach offers a practical and scalable solution for industrial motor drives, still having the inherent simplicity and cost-effectiveness of two-level VSIs.

Index Terms—Modulation, common-mode voltage, two-level voltage source inverter, carrier-phase-shift.

I. INTRODUCTION

Voltage source inverters (VSIs) play a significant role in industrial motor-drive applications [1]. Among the various VSI topologies, two-level VSIs (2L-VSIs) remain the preferred choice over more complex/advanced alternatives due to their simple structure, reduced component count, and high reliability [2]. Additionally, the emergence of wide-bandgap (WBG) devices like silicon carbide (SiC) and gallium nitride (GaN) has further enhanced the performance of 2L-VSIs by enabling higher switching frequencies, reducing losses, and improving overall efficiency [3]. Consequently, the combination of mature manufacturing processes for 2L-VSIs and the performance benefits of WBG devices provides a strong foundation, making the transition to newer topologies less attractive for many industrial applications. However, common-mode voltage

(CMV) issues, inherent to pulse-width modulation (PWM), are unavoidable in VSIs. The presence of CMV can induce common-mode currents that flow through the motor chassis or bearings [4], [5]. These currents accelerate bearing and insulation degradation, leading to fatigue failures and reduced system reliability [6]. Therefore, mitigating CMV is essential for enhancing the reliability and longevity of motor drive systems.

Various methods have been explored to mitigate CMV, including new topologies, filter designs, and innovative modulation strategies [7]. While multi-level or modified two-level topologies can reduce CMV, they often introduce increased cost and complexity [8]–[12]. In contrast, modulation-based solutions offer a practical alternative by reducing CMV without requiring hardware modifications, thus preserving system simplicity [13]–[19]. Therefore, this paper will focus on modulation-based techniques for CMV mitigation.

Several studies have explored carrier-phase shift techniques to mitigate common-mode voltage (CMV). The authors in [13] introduced interleaved carriers (0 , $-\frac{2\pi}{3}$, and $\frac{2\pi}{3}$) in three-phase SPWM, aiming to redistribute harmonics into positive, negative, and zero sequences. This approach was later extended to multiphase drives, including five-phase and both symmetrical and asymmetrical six-phase systems [14], as well as to three-level inverters [15]. Alternating carrier phase shift methods, which apply 0° or 180° shifts, were introduced for third harmonic injection SPWM, dividing the fundamental period into six segments and adjusting carrier phases accordingly [16]. A sector-based carrier phase shift approach was later employed in high-frequency SiC-based VSIs [17]. Further advancements include a sampling-time adaptive multi-carrier PWM algorithm, where four different carrier phase shift sets are evaluated at each sampling instant to minimize CMV [18]. More recently, a generalized multicarrier PWM technique was introduced, employing variable carrier phase shifts, optimizing both CMV and total harmonic distortion (THD) [19].

In all the aforementioned studies, the amount of carrier phase shifts either remains constant throughout a fundamental period or its polarity is adjusted based on the voltage sector which follows the desired reference voltage. While these methods effectively reduce the average CMV over a fundamental period, further improvement can be achieved by minimizing CMV at each switching interval. To address this, we propose an adaptive carrier phase shift approach for SPWM, where the phase shifts dynamically are adjusted within each switching interval. This method aims to further enhance overall CMV mitigation beyond the capabilities of conventional fixed- or sector-based phase-shift techniques.

The paper is organized as follows. Section II analyzes the SPWM and common mode voltages in the 2L-VSI. Section III explores the conventional interleaved multi-carrier SPWM. Section IV proposes and analyzes the adaptive carrier phase shift method to reduce the CMV. Section V discusses the influence of CMV frequency components. Section VI simulates the proposed modulation method and conventional methods to compare the performance regarding CMV reduction. Section VII gives the experimental validation of the proposed method. Section VIII concludes the paper and discusses potential future work.

II. ANALYTICAL INVESTIGATION OF COMMON MODE VOLTAGE FOR SPWM

The common mode voltage of a 2-Level 3-phase VSI, shown in Fig. 1, is calculated by

$$V_{CM} = \frac{1}{3}(V_A + V_B + V_C). \quad (1)$$

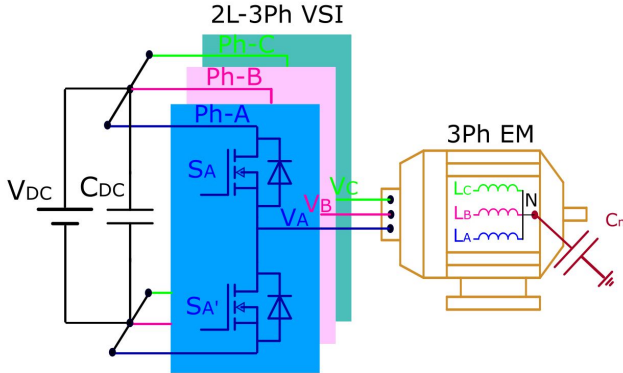


Fig. 1: The system structure of a 2-level 3-phase motor drive system.

The CMV leads to a common-mode current through motor shafts and bearings which can be modeled by the capacitor (C_n) connected between the neutral point and ground as given in Fig. 1. When conventional single-carrier SPWM is implemented, the phase-to-neutral voltages (V_{p-n}) in the

frequency domain can be calculated by using a Double Fourier series as:

$$V_{p-n} = \frac{V_{DC}}{2} + m_a \frac{V_{DC}}{2} \cos(\omega_R t + \theta_R) + \frac{2V_{DC}}{\pi} \sum_{m=1}^{\infty} J_0\left(m\frac{\pi}{2}m_a\right) \sin\left(m\frac{\pi}{2}\right) \cos\left(m(\omega_C t + \theta_C)\right) + \frac{2V_{DC}}{\pi} \sum_{m=1}^{\infty} \sum_{n=-\infty}^{\infty} \left[\frac{1}{m} J_n\left(m\frac{\pi}{2}m_a\right) \sin\left((m+n)\frac{\pi}{2}\right) \cos\left(m(\omega_C t + \theta_C) + n(\omega_R t + \theta_R)\right) \right], \quad (2)$$

where ω_R , ω_C , θ_R , θ_C , are angular frequencies and phases of reference and carrier signals, and J_0 , J_k are the zeroth and k^{th} order Bessel functions. The key components of the switching function, or leg voltage, include: the DC component, the baseband harmonics, the carrier harmonics, and the sideband harmonics. These harmonic components create positive, negative, and zero sequences regarding the phase shift of the carrier and reference signals. While positive and negative sequence voltage harmonics create differential-mode currents, zero-sequence voltage harmonics create common-mode currents. For the conventional SPWM with a single carrier signal, the common mode voltage harmonics can be calculated by using (1) and (2) as

$$V_{CM} = \frac{V_{DC}}{2} + \frac{2V_{DC}}{\pi} \sum_{\substack{m=1 \\ m \text{ is odd}}}^{\infty} \left[\frac{J_0\left(m\frac{\pi}{2}m_a\right) \sin\left(m\frac{\pi}{2}\right)}{\cos\left(m\omega_C t\right)} \right] + \frac{2V_{DC}}{\pi} \sum_{m=1}^{\infty} \sum_{\substack{n=-\infty \\ n \equiv 0 \pmod{3} \\ (m+n) \text{ is not even}}}^{\infty} \left[\frac{1}{m} J_n\left(m\frac{\pi}{2}m_a\right) \sin\left((m+n)\frac{\pi}{2}\right) \cos\left(m(\omega_C t) + n(\omega_R t)\right) \right]. \quad (3)$$

Therefore, the common-mode harmonics appear at odd carrier harmonics, at odd triplen harmonics near the even carrier harmonics, and at even triplen harmonics near the odd carrier harmonics. The sequences for the first carrier harmonics with sidebands are presented in Table I.

TABLE I: Frequency Components and Their Sequences Around First Carrier Harmonic.

Frequency	Phase-A	Phase-B	Phase-C	Sequence
f_0	0°	120°	-120°	Positive
$f_s - 6f_0$	0°	0°	0°	Zero
$f_s - 4f_0$	0°	-120°	120°	Negative
$f_s - 2f_0$	0°	120°	-120°	Positive
f_s	0°	0°	0°	Zero
$f_s + 2f_0$	0°	-120°	120°	Negative
$f_s + 4f_0$	0°	120°	0°	Positive
$f_s + 6f_0$	0°	0°	0°	Zero

Further, an example of an SPWM operation point with a single carrier signal is illustrated in Fig. 2, to show the common mode voltage.

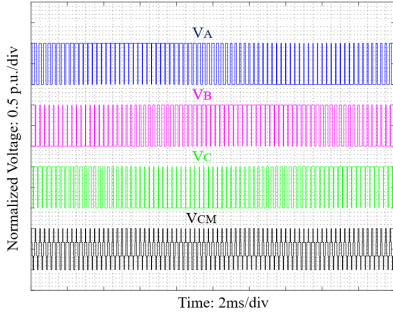


Fig. 2: The normalized common mode voltage for $m_a = 0.85$.

III. THE CONVENTIONAL INTERLEAVED MULTI-CARRIER SPWM

Using multi-carriers instead of a single carrier, the sequences of harmonics can be alternated by adjusting the phases of each carrier signal. The phases of the frequency component can be calculated using (2). The phases of the first carrier with sideband harmonics are given in Table II.

TABLE II: Frequency Components and Their Phases Around First Carrier Harmonic for Multi-Carrier Signals.

Frequency	Phase-A	Phase-B	Phase-C	Sequence
f_0	0°	120°	$-120^\circ + \phi_{CC}$	Positive
$f_s - 6f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$0^\circ + \phi_{CB} = -120^\circ$	$0^\circ + \phi_{CC} = 120^\circ$	Negative
$f_s - 4f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$-120^\circ + \phi_{CB} = 120^\circ$	$120^\circ + \phi_{CC} = -120^\circ$	Positive
$f_s - 2f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$120^\circ + \phi_{CB} = 0^\circ$	$-120^\circ + \phi_{CC} = 0^\circ$	Zero
f_s	$0^\circ + \phi_{CA} = 0^\circ$	$0^\circ + \phi_{CB} = -120^\circ$	$0^\circ + \phi_{CC} = 120^\circ$	Negative
$f_s + 2f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$-120^\circ + \phi_{CB} = 120^\circ$	$120^\circ + \phi_{CC} = -120^\circ$	Positive
$f_s + 4f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$120^\circ + \phi_{CB} = 0^\circ$	$0^\circ + \phi_{CC} = 0^\circ$	Zero
$f_s + 6f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$0^\circ + \phi_{CB} = -120^\circ$	$0^\circ + \phi_{CC} = 120^\circ$	Negative

$\phi_{CA} = 0$, $\phi_{CB} = -120^\circ$, and $\phi_{CC} = +120^\circ$.

Interleaving the carrier signals of the three phases is applied by shifting them by one-third of the switching period. Here, the main idea is to change the zero sequences with positive-negative sequences. The magnitude of the first carrier harmonic is greater than sidebands; therefore, altering the zero sequences to sidebands reduces the common mode voltage. Further, an example of an SPWM operation point with multi-carrier signals is given in Fig. 3, to show the change in common mode voltage.

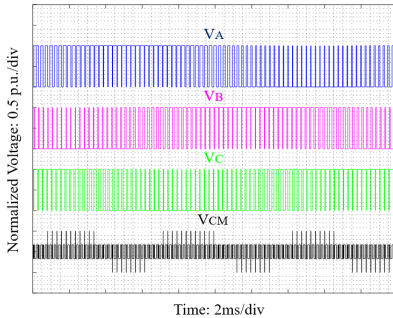


Fig. 3: The normalized common mode voltage for $m_a = 0.85$.

IV. THE PROPOSED ADAPTIVE CARRIER PHASE SHIFT METHOD

The proposed method introduces a variable carrier phase shift to achieve lower common-mode voltage (CMV) during each switching interval. Unlike conventional multi-carrier methods, where the phase shift is kept constant (or only the polarity is altered), this method adaptively adjusts the carrier phase shift to minimize CMV.

For each switching interval, the magnitude of the first switching component can be calculated using the Fourier series as follows:

$$\begin{aligned} V_A(t)^{\omega_c} &= \frac{2V_{DC}}{\pi} \sin(\pi D_A) \cos(\omega_c t + \phi_{CA}), \\ V_B(t)^{\omega_c} &= \frac{2V_{DC}}{\pi} \sin(\pi D_B) \cos(\omega_c t + \phi_{CB}), \\ V_C(t)^{\omega_c} &= \frac{2V_{DC}}{\pi} \sin(\pi D_C) \cos(\omega_c t + \phi_{CC}), \end{aligned} \quad (4)$$

where the duty cycles D_A , D_B , and D_C follow the reference sinusoidal waveforms as given by

$$\begin{aligned} D_A &= \frac{1 + m_a \cos(\omega_R t)}{2}, \\ D_B &= \frac{1 + m_a \cos(\omega_R t + 120^\circ)}{2}, \\ D_C &= \frac{1 + m_a \cos(\omega_R t - 120^\circ)}{2}. \end{aligned} \quad (5)$$

The common-mode voltage for the first carrier harmonic is then calculated within a switching interval as

$$V_{CM}^{\omega_c}(t) = \frac{1}{3} [V_A(t)^{\omega_c} + V_B(t)^{\omega_c} + V_C(t)^{\omega_c}]. \quad (6)$$

Thus, we have three signals ($V_A(t)^{\omega_c}$, $V_B(t)^{\omega_c}$, and $V_C(t)^{\omega_c}$) with the same frequency but different magnitudes and phases. This allows us to analyze CMV in the phasor domain, as illustrated in Fig. 4.

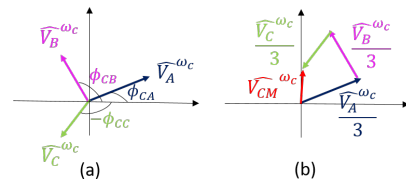


Fig. 4: Phasor domain analysis of the CMV in the first switching harmonic. a) The phase voltages. b) The CMV.

The goal is to minimize $V_{CM}^{\omega_c}$. The three voltage vectors have magnitudes that vary with time and modulation index, as shown in Fig. 5. If these three vectors form a triangle, their sum can be minimized to zero. This condition can be verified using the triangle inequality. If the vectors do not form a triangle, the minimum CMV can be achieved by aligning the smaller vectors in opposition to the larger vector. In the first case (vectors forming a triangle), the carrier phase shifts can be

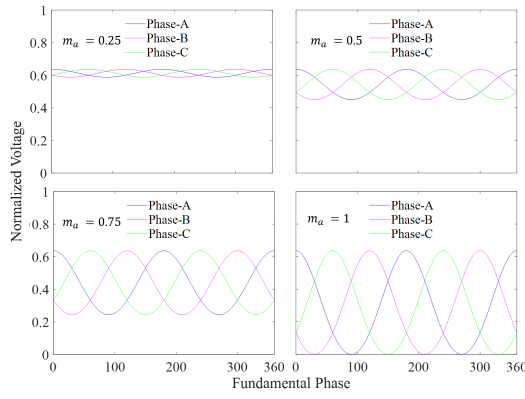


Fig. 5: Magnitude of vectors over a fundamental period for different modulation indexes.

determined using the cosine and sine theorems. In the second case (vectors that do not form a triangle), the phases of the smaller vectors are set to 180° relative to the larger vector, effectively canceling its contribution to the CMV. The cases are illustrated in Fig. 6.

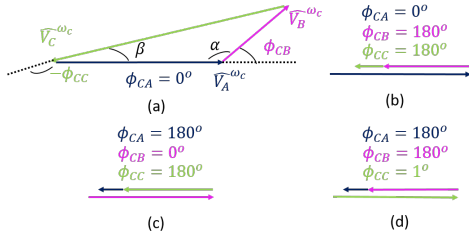


Fig. 6: Determination of the carrier phase shifts in phasor domain analysis. a) Triangle formation. b) Phase-A voltage vector is bigger than other vectors. c) Phase-B voltage vector is bigger than other vectors. d) Phase-C voltage vector is bigger than other vectors.

To make the calculation easy, ϕ_{CA} is taken as zero in triangle formation, and the other two carrier phase shifts can be calculated using the Algorithm 1. Further, an example of an SPWM operation point with the adaptive carrier phase shift method is shown in Fig. 7, to show the change in common mode voltage.

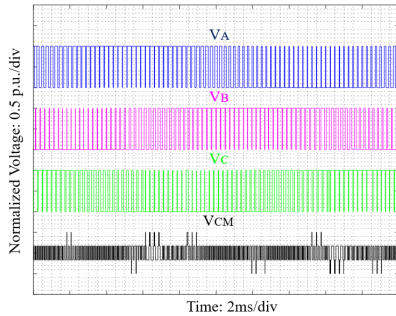


Fig. 7: The normalized common mode voltage for $m_a = 0.85$.

Algorithm 1 Determination of the Carrier Phase Shifts

- 1: **if** $|\hat{V}_A^{\omega_c} - \hat{V}_B^{\omega_c}| < \hat{V}_C^{\omega_c} < |\hat{V}_A^{\omega_c} + \hat{V}_B^{\omega_c}|$ **then**
- 2: Take $\varphi_{CA} = 0^\circ$:
- 3: Apply cosine theorem to find φ_{CB} :

$$(\hat{V}_A^{\omega_c})^2 + (\hat{V}_B^{\omega_c})^2 - 2\hat{V}_A^{\omega_c}\hat{V}_B^{\omega_c}\cos(\alpha) = (\hat{V}_C^{\omega_c})^2$$

$$\cos[\alpha] = \frac{(\hat{V}_A^{\omega_c})^2 + (\hat{V}_B^{\omega_c})^2 - (\hat{V}_C^{\omega_c})^2}{2\hat{V}_A^{\omega_c}\hat{V}_B^{\omega_c}}$$

$$\alpha = \arccos\left[\frac{(\hat{V}_A^{\omega_c})^2 + (\hat{V}_B^{\omega_c})^2 - (\hat{V}_C^{\omega_c})^2}{2\hat{V}_A^{\omega_c}\hat{V}_B^{\omega_c}}\right]$$

$$\varphi_{CB} = 180^\circ - \alpha$$
- 4: Apply sine theorem to find φ_{CC} :

$$\frac{\sin[\beta]}{\hat{V}_B^{\omega_c}} = \frac{\sin[\alpha]}{\hat{V}_C^{\omega_c}}$$

$$\sin[\beta] = \frac{\hat{V}_B^{\omega_c}\sin[\alpha]}{\hat{V}_C^{\omega_c}}$$

$$\beta = \arcsin\left[\frac{\hat{V}_B^{\omega_c}\sin[\alpha]}{\hat{V}_C^{\omega_c}}\right]$$

$$\varphi_{CC} = \beta - 180^\circ$$
- 5: **else**
- 6: **if** $[\hat{V}_A^{\omega_c} > \hat{V}_B^{\omega_c}]$ **and** $[\hat{V}_A^{\omega_c} > \hat{V}_C^{\omega_c}]$ **then**
- 7: $\varphi_{CA} = 0^\circ$, $\varphi_{CB} = 180^\circ$ and $\varphi_{CC} = 180^\circ$
- 8: **else if** $[\hat{V}_B^{\omega_c} > \hat{V}_A^{\omega_c}]$ **and** $[\hat{V}_B^{\omega_c} > \hat{V}_C^{\omega_c}]$ **then**
- 9: $\varphi_{CA} = 180^\circ$, $\varphi_{CB} = 0^\circ$ and $\varphi_{CC} = 180^\circ$
- 10: **else if** $[\hat{V}_C^{\omega_c} > \hat{V}_A^{\omega_c}]$ **and** $[\hat{V}_C^{\omega_c} > \hat{V}_B^{\omega_c}]$ **then**
- 11: $\varphi_{CA} = 180^\circ$, $\varphi_{CB} = 180^\circ$ and $\varphi_{CC} = 0^\circ$
- 12: **end if**
- 13: **end if**

V. DISCUSSION

Bearing currents originate from different physical mechanisms, such as capacitive bearing currents, electric discharge machining (EDM) currents, and circulating bearing currents [7]. For capacitive bearing currents, high-frequency components in the CMV are critical because the current amplifies at higher frequencies. This means that the CMV related to voltage slope (dv/dt) is more important than the CMV related to the switching pattern. However, the switching pattern still has a significant effect on EDM currents, since beyond a certain threshold voltage, the lubricant in the bearings undergoes electrical breakdown. This highlights the importance of reducing the total magnitude of the CMV.

VI. SIMULATION RESULTS

The simulation study compares the conventional interleaved SPWM and the proposed adaptive multi-carrier SPWM. The modulation indices of $m_a = 0.4, 0.8, 1$ are selected. Then, fast Fourier transforms (FFTs) of these CMVs are calculated to compare their performance. The CMVs and their FFTs are shown in Fig. 8.

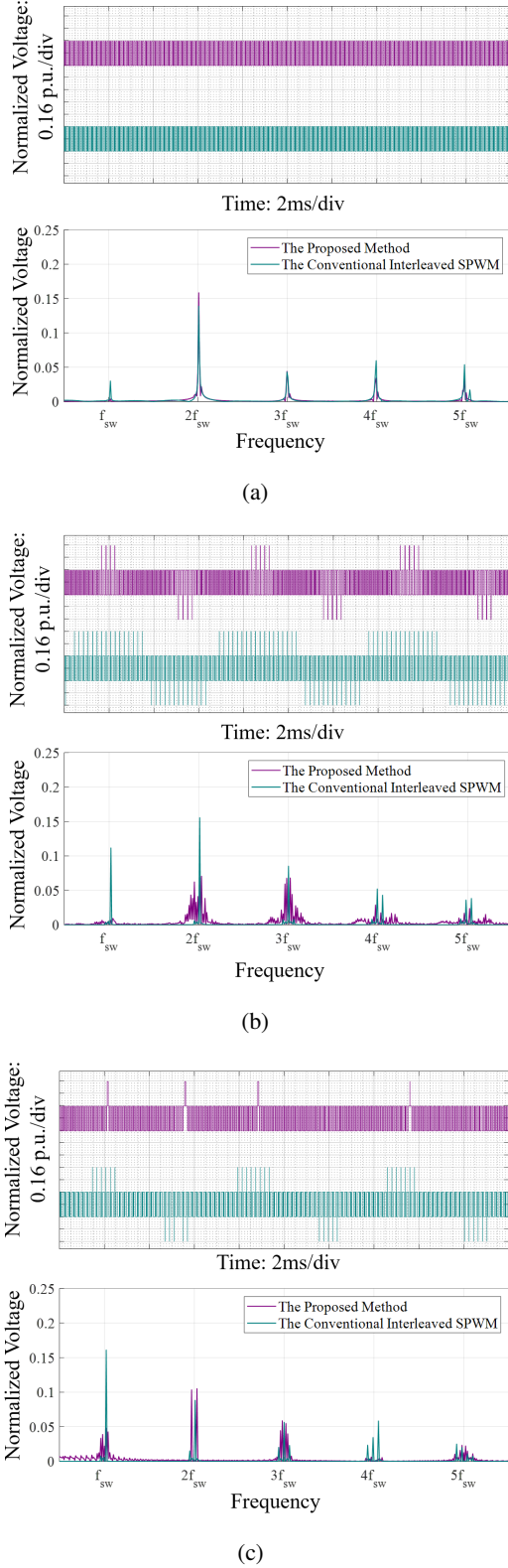


Fig. 8: The CMVs and their FFTs for three modulation indices. a) $m_a = 0.4$. b) $m_a = 0.8$. c) $m_a = 1$.

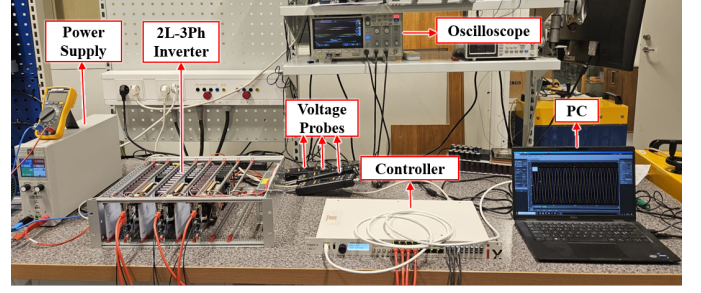


Fig. 9: Experimental setup of 2-level 3-phase inverter system.

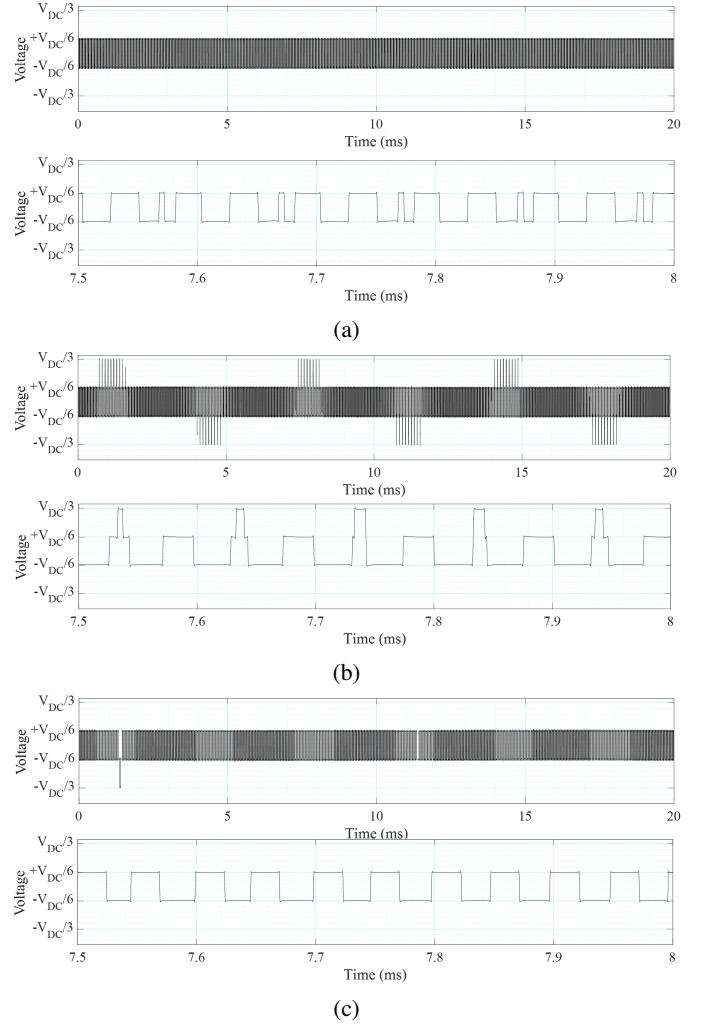


Fig. 10: The CMVs for three modulation indices (at fundamental frequency of $50Hz$. and switching frequency of $10kHz$). a) $m_a = 0.4$. b) $m_a = 0.8$. c) $m_a = 1$.

VII. EXPERIMENTAL VALIDATION

An experimental setup, as illustrated in Fig. 9, has been established to validate the proposed methodology. The setup uses Imperix SiC power module PEB8024 half-bridge configurations to achieve a 2-Level 3-Phase inverter. The Imperix B-Box RCP controller is implemented to generate and execute

the proposed modulation strategy, enabling real-time control. The experimental waveforms for the same modulation index with the simulation results were taken as shown in Fig. 10. It is observed that the experimental results are in good agreement with the simulated results.

VIII. CONCLUSION

This study proposes an adaptive carrier phase shift technique for sinusoidal pulse width modulation (SPWM) in two-level voltage source inverters (VSIs) to effectively reduce common-mode voltage (CMV). Unlike conventional methods that use fixed or sector-based phase shifts, the proposed approach dynamically adjusts the carrier phase shift within each switching interval, leading to a significant reduction in instantaneous CMV. The analytical investigation and experimental validation demonstrated that the proposed method enhances CMV mitigation, thereby improving system reliability and electromagnetic interference (EMI) performance. The simplicity of the two-level VSI structure is maintained, making this approach a practical solution for industrial motor-drive applications. Future work could explore applying this technique in multi-phase or multi-level inverters and further optimize for different operating conditions.

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